



PRODUCTS

- Air Cooled Vibration Systems
- DSA Power Amplifiers
- Vibration Controllers
- Bump/Shock Test Machine
- Vibration Application for Elastomers & Transmission Line Dampers

SERVICES

- Engineer & Customer Training
- Maintenance Contract Services
- Refurbished System Supply
- Spare Part Supply
- Armature Coil Rewinding
- Replacement of Power Amplifiers
- Application Support

TYPICAL SYSTEM APPLICATIONS

- Avionic Testing
- Automotive Component Testing
- Electronic Component/Assembly Testing
- Stress Screening
- Sine, Random & Transient Testing
- Natural Frequency Search
- Life Time Simulation
- Product Design Validation
- Product Qualification
- Transport & Packaging
- Halt/HASS Simulation



VIBRATION SYSTEM CONFIGURATION

TABLE-1

Model				S001/P05	S002/P02	S005/P05	S010/P1	S020/P2	S040/P4	S060/P6
System Parameters	Frequency Range Hz			5~4000	5~4000	5~4000	5~4000	5~4000	5~3500	5~3000
	Rated Force	Sine	(Kgf)	10	20	50	100	200	400	600
			(KN)	0.1	0.2	0.5	1.0	2.0	4.0	6.0
		Random	(Kgf)	7	14	38	74	148	283	426
			(KN)	.07	0.2	0.4	0.7	1.4	2.8	4.2
	Max. Acceleration (g)			60	70	70	80	80	80	80
	Max. Velocity (m/s)			1.4	1.5	1.5	1.6	1.8	1.8	1.6
	Max. Disp. (mm, p-p)			6.0	10	12	18	20	20	25
	Max. Loading weight (Kg)			1.5	3.0	6.0	40	60	120	150
Utility Power (KVA)			0.2	0.5	1.0	2.0	3.0	4.5	9.0	
Shaker	Model			AEV001	AEV002	AEV005	AEV010	AEV020	AEV040	AEV060
	Mass of Armature (Kg)			0.4	0.6	0.8	2.0	3.0	6.0	7.5
	Armature diameter (ϕmm)			30	50	100	125	140	160	160
Power Amplifier	Model			DSA05K	DSA05K	DSA1K	DSA2K	DSA2K	DSA4K	DSA6K
	Max. Output (KVA)			0.1	0.2	0.5	1.0	2.0	4.0	6.0
	Technology			PWM	PWM	PWM	PWM	PWM	PWM	PWM
Controller	Vibration Controller			PC Based Digital Vibration Shaker Controller (USB Interface and Win XP based Sine, Random, Shock Software)						
Cooling System	Model			N/R	Optional	ACS002	ACS002	ACS002	ACS004	ACS006
	Type of Cooling				Air cooled	Air cooled	Air cooled	Air cooled	Air cooled	Air cooled

TABLE-2

Model				S100/P9	S130/P12	S200/P20	S250/P30	S300/P36	S350/P42	S400/P48
System Parameters	Frequency Range Hz			5~3000	5~2500	5~2500	5~2500	5~2500	5~2500	5~2500
	Rated Force	Sine	(Kg ^f)	1000	1300	2000	2500	3000	3500	4000
			(KN)	10.0	13.0	20.0	25.0	30.0	35.0	40.0
		Random	(Kg ^f)	721	940	1431	1770	2130	2480	2830
			(KN)	7.2	9.4	14.3	17.7	21.3	24.8	28.0
	Max. Acceleration (g)			90	90	90	80	90	85	90
	Max. Velocity (m/s)			1.6	1.4	1.4	1.4	1.4	1.4	1.4
	Max. Disp. (mm, p-p)			38	38	38	45	45	51	51
	Max. Loading weight (Kg)			250	250	400	450	500	550	550
System Power Consumption (KVA)			15	20	32	40	48	60	72	
Shaker	Model			AEV100	AEV130	AEV200	AEV250	AEV300	AEV350	AEV400
	Mass of Armature (Kg)			14	32	32	34	40	44	46
	Armature diameter (ϕmm)			240	300	300	300	360	410	410
Power Amplifier	Model			DSA10K	DSA12K	DSA24K	DSA32K	DSA36K	DSA40K	DSA48K
	Max. Output (KVA)			10	12	24	32	36	40	48
	Technology			PWM	PWM	PWM	PWM	PWM	PWM	PWM
Controller	Vibration Controller			PC Based Digital Vibration Shaker Controller (USB Interface and Win XP based Sine, Random, Shock Software)						
Cooling System	Model			ACS009	ACS010	ACS010	ACS011	ACS011	ACS012	ACS012
	Type of Cooling			Air cooled	Air cooled	Air cooled	Air cooled	Air cooled	Air cooled	Air cooled

Note : 1. Shaker can be modified as per the Customer Application Requirement.
2. For higher Random force re-rated amplifier and shaker are also offered.

PAY LOAD SUPPORT GUIDELINES FOR SHAKERS

Combo Base Horizontal Slip Table

CB HST Model	Table Size (mmxmm)	Mass (Kg) Al	Shaker Model
HST 3030	300×300	19	AEV 010 to 020
HST 4060	400×600	30	AEV 040 to 100
HST 6060	600×600	40	AEV 040 to 100
HST 6080	600×800	55	AEV 060 to 100
HST 60100	600×1000	70	AEV 130 to 200
HST 80100	800×1000	75	AEV 200 to 400
HST 100100	1000×1000	120	AEV 200 to 400

Vertical Load Support

Model	Size (mm)	Mass (Kg) Al	Mg	Shaker Model
VLS 30	300	16	12	AEV 010 to 020
VLS 40	400	25	19	AEV 020 to 040
VLS 50	500	30	22	AEV 040 to 100
VLS 55	550	35	25.5	AEV 040 to 100
VLS 60	600	40	29	AEV 040 to 100
VLS 80	800	60	45	AEV 130 to 200
VLS 100	1000	110	83	AEV 200 to 400

Square Head Expander

Model	Table Size (mm)	Mass (Kg) Al	Mg	Shaker Model
SHE 1010	100×100	1.2	0.9	AEV 005
SHE 2020	200×200	5	3.5	AEV 020
SHE 3030	300×300	12.5	9.5	AEV 060
SHE 4040	400×400	17	12.5	AEV 130
SHE 6060	600×600	22.5	17	AEV 130
SHE 60100	600×1000	35	26	AEV 200
SHE 8080	800×800	60	45	AEV 250
SHE 100100	1000×1000	95	71	AEV 350

Circular Head Expander

Model	Size (mm)	Mass (Kg) Al	Mg	Shaker Model
CHE 10	100	1.2	0.9	AEV 010 to 020
CHE 20	200	4.5	3.2	AEV 020 to 040
CHE 30	300	10	7.5	AEV 040 to 100
CHE 40	400	15	11	AEV 040 to 100
CHE 50	500	21	16	AEV 040 to 100
CHE 60	600	26	20	AEV 130 to 200
CHE 80	800	50	37	AEV 200 to 400
CHE 100	1000	56	83	AEV 130 to 200

NOTE : Special Pay load support having different mounting hole pattern and thread size can be designed for Specific applications.

SYSTEM RATINGS

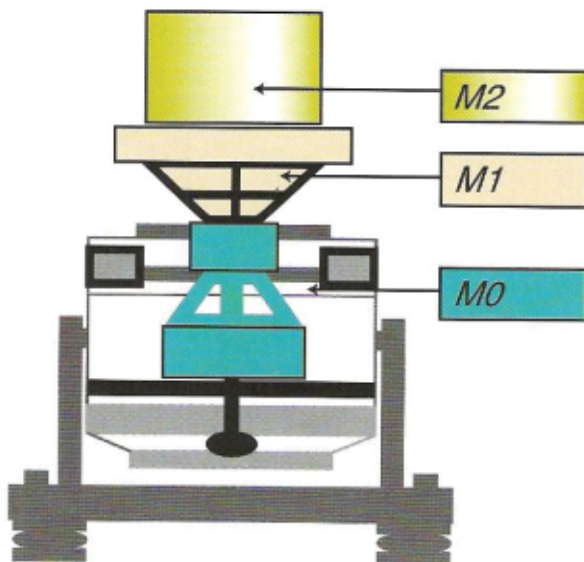
Electro Dynamic Vibration Test Systems are used to satisfy a wide variety of vibration test requirement. Sine, Random and Shock testing are considered the three basic test environments and are reflected in the standard ratings for all Apson vibration test systems. All vibration test systems are rated in peak force for sine and shock and in rms force for random. All systems are rated to 2,000 Hz. Some systems will deliver full performance to 3,000 Hz. All systems will operate to 3,000 Hz or beyond at reduced performance.

SINE : Sine Performance is defined by an envelope of maximum displacement, velocity, and acceleration as a function of frequency for a given Amplifier and shaker combination.

RANDOM : Random performance is based upon a grms level over a specific frequency range. In order to ensure uniformity a specific set of conditions must be used to eliminate test variables. These conditions include a flat spectral density (g^2 / Hz) from 20 Hz to 2,000Hz and a heavy, non-resonant load. This method of rating provides consistency and exceeds the requirements of ISO 5344, if a shaped spectrum is used, the random force rating may vary.

SHOCK : Shock performance is defined for various pulse waveforms is half sine, saw tooth. A family of curves is provided for each pulse waveform based upon a given pulse duration and peak acceleration.

RATED FOREC CALCULATION :



CALCULATION :

1.	SINE FORCE (p-p) $F = M \times A = (M_0 + M_1 + M_2) \times a \times 1.2$
	<i>M0 : Armature mass (kgw)</i>
	<i>M1 : Auxiliary Table or Fixture (kgw)</i>
	<i>M2 : Specimen (kgw)</i>
	<i>a : Vibration acceleration (g)</i>
	<i>1.2 : Safety Factor</i>
2.	RANDOM FORCE (RMS) $= \text{SINE FORCE} \times \sqrt{2}$

$$\Sigma M = M_0 + M_1 + M_2$$

Note : Because of continuous product improvement, specifications are subject to change without advance notice. Latest status will always be indicated in our quotation.



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